

Course Code	Course Title	L	T	P	C
BECM302E	Augmented and Virtual Reality	3	0	2	4
Pre-requisite	NIL	Syllabus version			
		1.0			
Course Objectives					
The objective of this course is to					
1. To introduce the fundamentals of virtual reality systems, including geometric modelling, transformations, graphical rendering, optics, and human vision.					
2. To build a personal VR or AR application for their portfolio					
3. To develop Virtual and Augmented Reality industry committed innovative technologies that can be used in the development and production of immersive environments in the fields of entertainment, education, training, medical and industrial innovation.					
Course Outcomes					
After successfully completing the course, the student should be able to					
1. Understand the design, types and the principles of immersive technologies.					
2. Design, create, and integrate 3D art elements into a virtual reality environment.					
3. Applying motion tracking techniques, visual, and interactive elements into a virtual reality environment.					
4. Design, create, and calibrate the platform, motion tracking and lighting in augmented reality environment.					
5. Develop techniques to work on multiple platforms, including PC, mobile devices, etc.,					
6. Geometric modeling and using immersive effects of visual and audio assets to AR/VR experiences.					
7. Adapt VR/AR designs to meet requirements of the current marketplace.					
8. Analyze and design prototypes for serious applications.					
Module:1	Design and Art Across Digital Realities	5 hours			
Introduction – How human interacts with computers – Modalities through Ages – Types of Common Human Computer Interaction (HCI) Modalities: Visual, Auditory and Physical - New Modalities – Note on Hand Tracking and Hand Pose Recognition – Voice, Hands and Hardware Inputs over Next Generation – Sensory Design – Sensory Principles					
Module:2	Introduction to Virtual Reality	6 hours			
A Brief History of Virtual Reality - The five Classic Components of a VR System - Reality, Virtuality and Immersion -Virtual Reality for Art – VR for Animation – 3D Art Optimization –Ideal Solution – Topology – Baking – Draw Calls – VR Tools to Create 3D Art – Acquiring 3D models – Creating 3D models.					
Module:3	Motion tracking, navigation and controllers in VR	6 hours			
Position and Motion Trackers - Magnetic, Mechanical and Ultrasonic Trackers - Navigation and Manipulation Interfaces - Three-Dimensional Probes and Controllers - Data Gloves and Gesture Interfaces - Human Perception and Cognition – Visual System – Effects of VR Simulations on Users - Cyber-sickness, before and now - Guidelines for Proper VR Usage - User Centered Design, User Experience and an Ethical Code of Conduct					
Module:4	Introduction to Augmented Reality	6 hours			

History of AR – Selection of AR Platform– Integrating Hardware and Software – Optical & Inertial Calibration – Tracking – AR Computer Vision – Mapping – Platforms – Lightings		
Module:5	Creating Cross-Platform Augmented Reality and Virtual Reality	6 hours
Cross Platform Game Engines – Understanding 3D graphics – Virtual Camera – Degree of Freedom – Virtual Reality Toolkit – Best Practices		
Module:6	Modeling Interactivity Sound in Immersive Environments	6 hours
Geometric Modeling - Kinematics Modeling - Physical Modeling - Behavior Modeling - Model Management – Handling locomotion: Locomotion in AR/VR, Effective use of audio: Audio in AR/VR, Common Interaction Paradigms: Inventory for AR, Augmented Reality Raycasts		
Module:7	Use Cases in the field of VR/AR	6 hours
Health: VR/AR Health Technology Application Design, Sports: Five Key Principles of AR and VR for Sports, Training: Enterprise Training, Soft Skills Training, Voice Recognition		
Module:8	Recent Trends in AR/VR	4 hours
Intelligent system design using the AI enabled AR/VR		
	Total Lecture hours:	45 hours
Text Book(s)		
1.	Erin Pangilinan, Steve Lukas and Vasanth Mohan, “Creating Augmented and Virtual Realities Theory & Practice for Next-Generation Spatial Computing”, O’REILLY 2019	
2.	Coiffet P, Burdea GC. Virtual reality technology. John Wiley & Sons; 2017 Nov 1.	
Reference Books		
1.	Tom Dieck MC, Jung TH, Loureiro SM, editors. Augmented Reality and Virtual Reality: New Trends in Immersive Technology. Springer Nature; 2021.	
2	Jason Jerald. 2015. The VR Book: Human-Centered Design for Virtual Reality. Association for Computing Machinery and Morgan & Claypool Publishers	
3	Tony Parisi. 2015. Learning Virtual Reality ISBN: 9781491922828	
4	Michael Madary and Thomas K. Metzinger. 2016. Real Virtuality: A Code of Ethical Conduct. Recommendations for Good Scientific Practice and the Consumers of VR- Technology. Frontiers in Robotics and AI 3, February: 1–23. http://doi.org/10.3389/frobt.2016.00003	
Mode of Evaluation: Continuous Assessment Test, Digital Assignment, Quiz and Final Assessment Test		
Indicative Experiments		
1.	VR/AR engines - UNITY Basics/ Google ARCore/ Vuforia, etc.,	3 hours
2.	3D environment creation using UNITY	3 hours
3.	2D environment creation using UNITY	3 hours
4.	Configure project settings according to VR platform requirements	3 hours
5.	Creating a standalone VR application	3 hours

6.	Configure project settings according to AR platform requirements	3 hours
7.	Creating a standalone AR application	3 hours
8.	Set up ARCore development environment	3 hours
9.	Determine the components needed for a user to physically manipulate objects in a VR environment	3 hours
10.	Understand types of head tracking and degrees of movement in VR equipment	3 hours
Total Laboratory Hours		30 hours
Mode of Assessment: Continuous Assessment and FAT		
Recommended by Board of Studies		28-02-2023
Approved by Academic Council		No. 69 Date 16-03-2023